

A2 Warm Up P-6 Binomial Probability

$$P(r) = \binom{n}{r} (p)^r (q)^{n-r}$$

Trials # of Successes prob success prob fail

1. The Hiking Club plans to go camping in a State park where the probability of rain on any given day is 0.22. What is the probability that it will rain on exactly one of the three days they are there? Round your answer to the nearest thousandth.

$$p = .22 \quad q = 1 - .22 = .78$$

$$r = 1 \quad n = 3$$

$$P(1) = {}_3C_1 (.22)^1 (.78)^2$$

$$= .401544$$

$$= \boxed{.402}$$

2. If a fair coin is tossed 9 times, what is the probability, to the nearest thousandth, of getting exactly 2 heads? Heads or Tails

$$p = .5 \quad q = .5 \quad n = 9 \quad r = 2$$

$$P(2) = {}_9C_2 (.5)^2 (.5)^7$$

$$= .0703125$$

$$= \boxed{.070}$$

3. During a single day at radio station WMZH, the probability that a particular song is played is 0.45. What is the probability that this song will be played on at least 1 days out of 4 days? Round your answer to the nearest thousandth.

$$n = 4 \quad \text{"at least 1"} \Rightarrow r = 1$$

$$r = 2$$

$$r = 3$$

$$r = 4$$

$$p = .45 \quad q = .55$$

$$P(r \geq 1) = {}_4C_1 (.45)^1 (.55)^3 +$$

$${}_4C_2 (.45)^2 (.55)^2 +$$

$${}_4C_3 (.45)^3 (.55)^1 =$$

$${}_4C_4 (.45)^4 (.55)^0 =$$

$$= .299475 + .3675375 + .200475$$

$$+ .04100625 = \boxed{.908}$$

★
FASTER:

$$P(r \neq 0) = 1 - P(r = 0) = 1 - {}_4C_0 (.45)^0 (.55)^4$$

4. Mr. and Mrs. Doran have a genetic history such that the probability that a child being born to them with a certain trait is 60%. If they have seven children, what is the probability that at most two of their seven children will have that trait? Round your answer to the nearest thousandth.

$$p = .60 \quad n = 7 \quad r = 0, 1, 2$$

$$q = .40$$

$$P(r \leq 2) = {}_7C_0 (.60)^0 (.40)^7 +$$

$${}_7C_1 (.60)^1 (.40)^6 +$$

$${}_7C_2 (.60)^2 (.40)^5 =$$

$$= .0016384 + .0172032 +$$

$$.0774144$$

$$= .096256$$

$$= \boxed{.096}$$